ĎALEKOHĽAD

Rozsah: 1 cvičenie – 4 hodiny

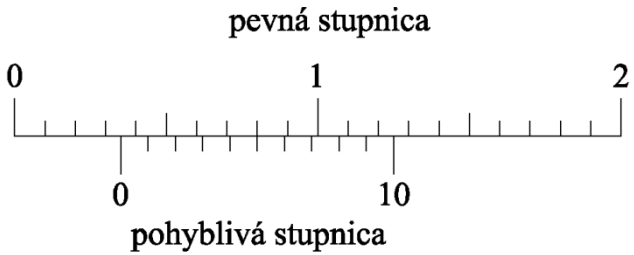
Zadanie

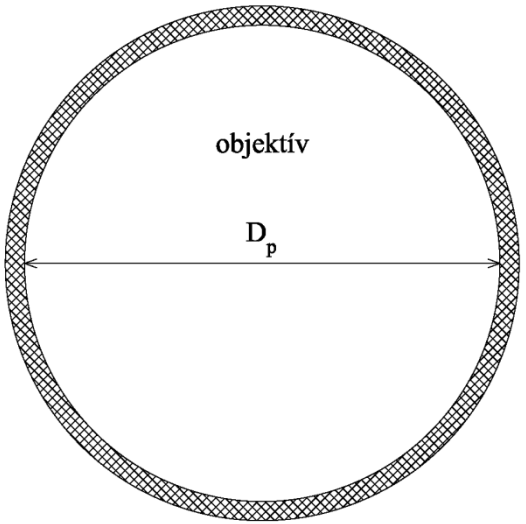
Určenie výkonnosti ďalekohľadu teodolitov Nivel System DT-2 a Zeiss Theo 020A:

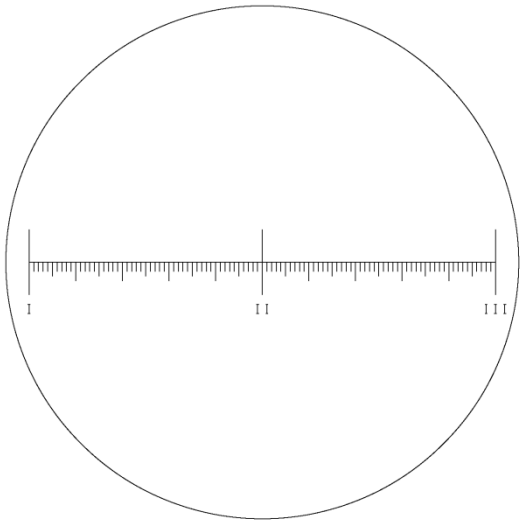
1. zväčšenie (5-krát),
2. zorné pole (5-krát na troch dĺžkach 10, 12 a 14 m),
3. rozlišovacia schopnosť (5-krát na troch dĺžkach 10, 12 a 14 m),
4. geometrická svetelnosť.

Zväčšenie

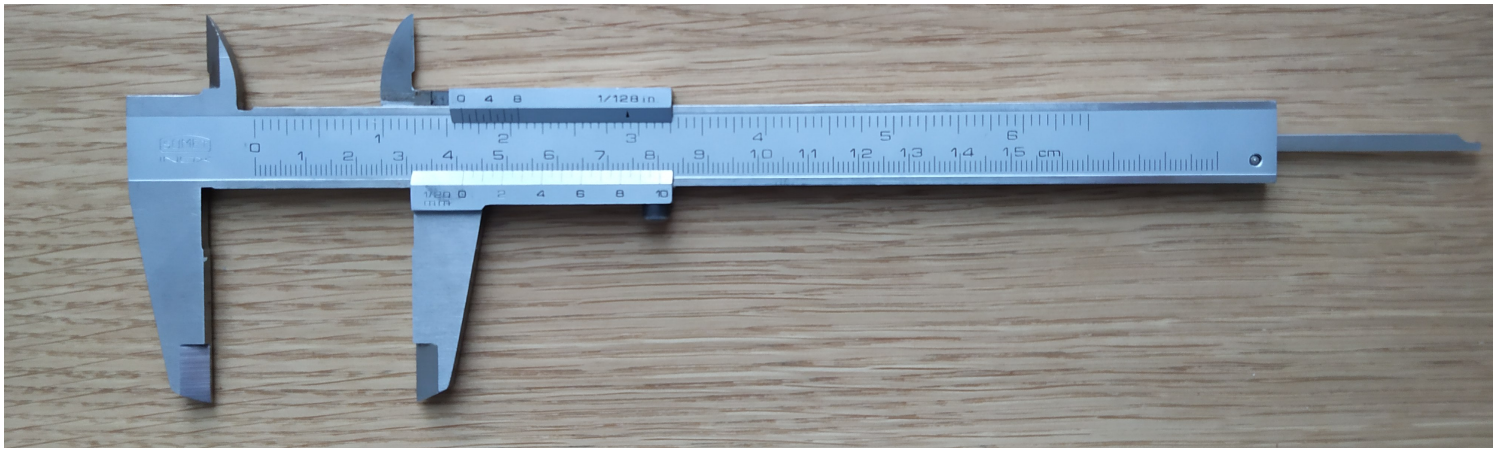
$$\Gamma = \frac{D_p}{d_p}$$

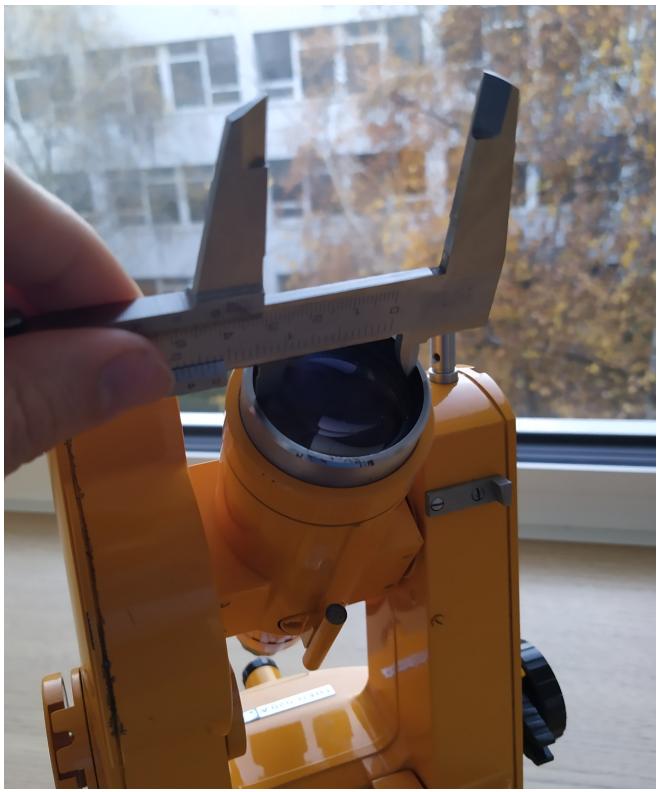


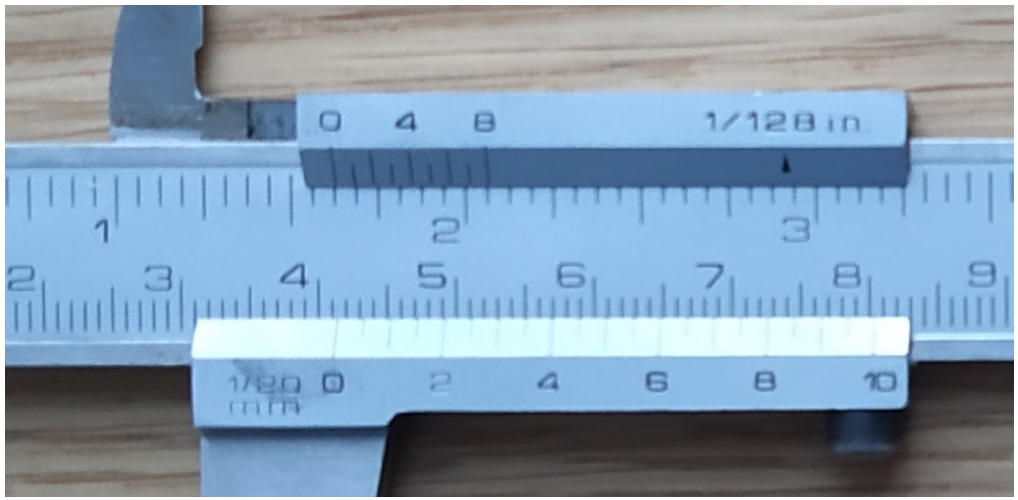




n	D_p [mm]	d_p [mm]	Γ
1	35,2	1,5	23,5
2	35,8	1,5	23,9
3	35,6	1,6	22,3
4	35,5	1,6	22,2
5	36,1	1,5	24,0

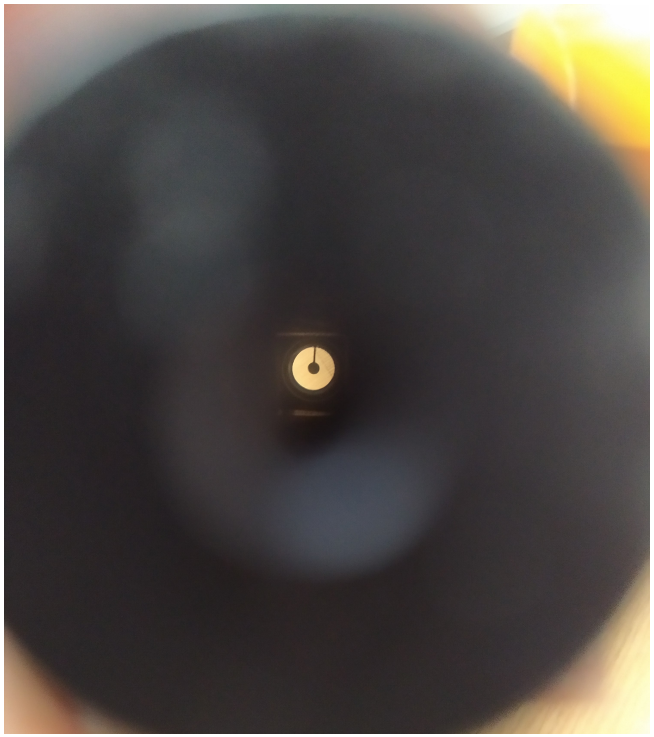

0 4 8 1/128 in
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 cm
1/20 0 2 4 6 8 10
mm

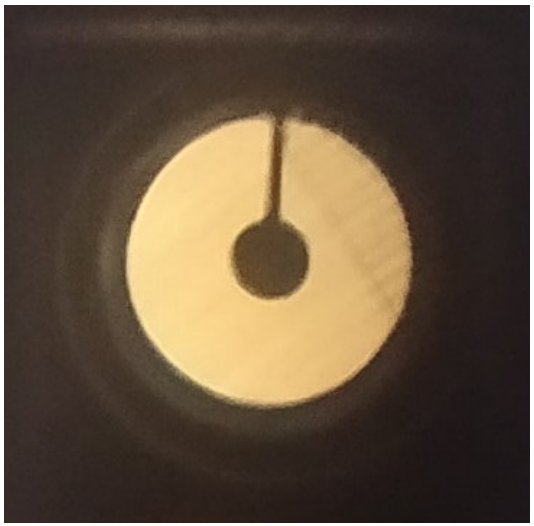



0 4 8 1/128 in
1 2 3
2 3 4 5 6 7 8 9
1/20 0 2 4 6 8 10
mm


15
10
5

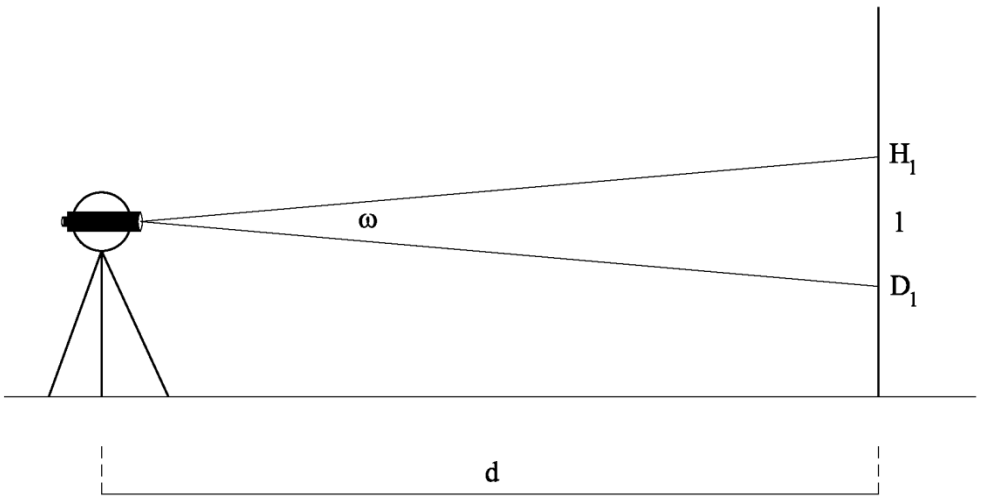

THEO 020 A

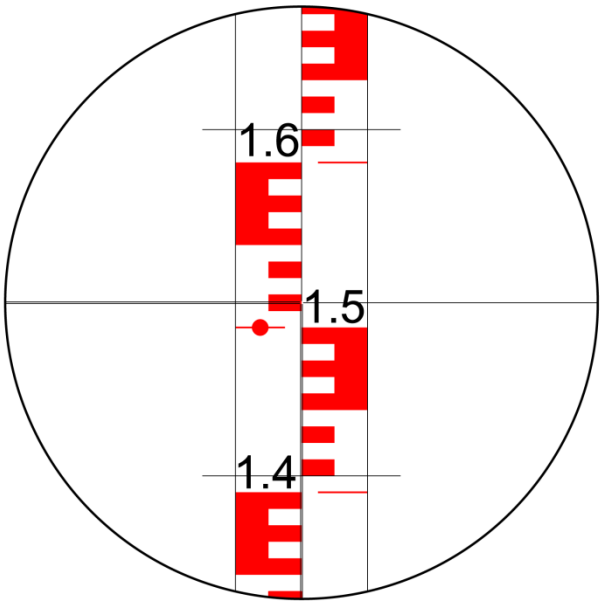




Zorné pole

$$\omega' = \frac{l}{d} \cdot \rho' \qquad \rho' = \frac{180 \cdot 60}{\pi}$$

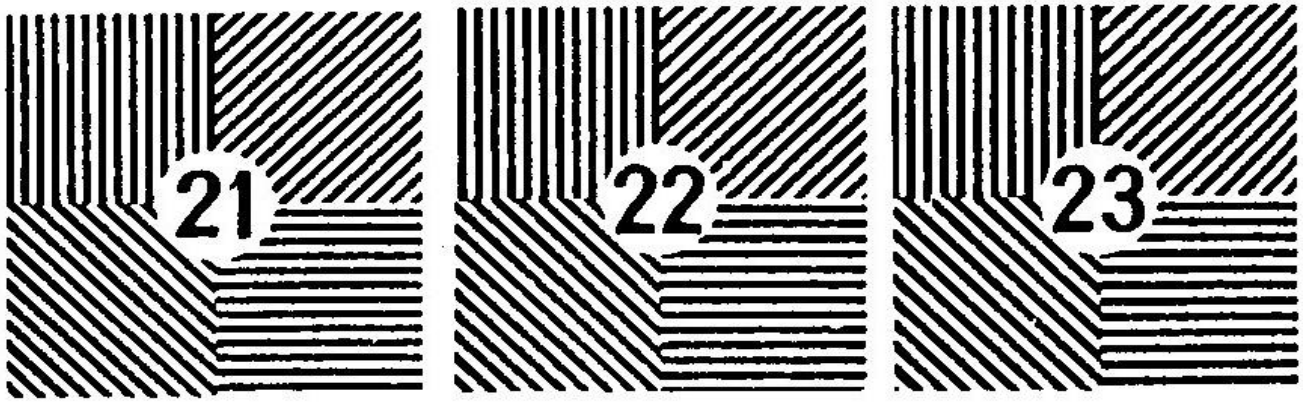




d [m]	n	D_l [m]	H_l [m]	l [m]	ω'
10	1	1,550	1,648	0,098	34
	2	1,600	1,699	0,099	34
	3	1,650	1,749	0,099	34
	4	1,700	1,799	0,099	34
	5	1,750	1,849	0,099	34
12	1	1,550	1,669	0,119	34
	2	1,600	1,721	0,121	35
	3	1,650	1,772	0,122	35
	4	1,700	1,831	0,131	38
	5	1,750	1,881	0,131	38
14	1	1,600	1,741	0,141	35
	2	1,650	1,789	0,139	34
	3	1,700	1,839	0,139	34
	4	1,750	1,891	0,141	35
	5	1,800	1,936	0,136	33

Rozlišovacia schopnosť

$$\psi'' = \frac{10 \cdot n}{d}$$



d [m]	10	12	14
n	6	8	8
ψ''	6,0	6,7	5,7

Geometrická svetelnosť

$$S = \frac{D_p^2}{d_o^2 \cdot \Gamma^2} = \frac{d_p^2}{d_o^2}$$

Výsledky

Prístroj	Zeiss Theo 020A	Nivel System DT-2
Γ		
ω'		
ψ''		
S		

Obsah elaborátu

1. Technická správa s postupom určenia parametrov
2. Príloha A: Výpočtový protokol
3. Príloha B: Chod lúčov Keplerovým ďalekohľadom